

***TESS* photometry of the Of?p stars HD 148937 and LMC 164-2**

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Introduction:



Of?p Stars:

- All known Galactic Of?p stars are magnetic.
- Magnetic O stars exhibit photometric variability due to their rotating magnetospheres.
- Six known Of?p stars in the Magellanic Clouds that show rotational modulation.
- About a dozen confirmed magnetic O stars.

Goal: Utilize *TESS* observations to identify new short period magnetic O star candidates (MOBSTER).

TESS Photometry:

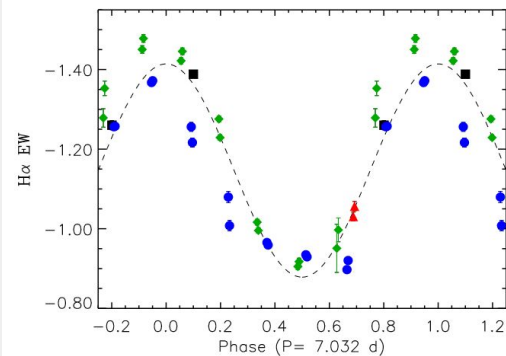
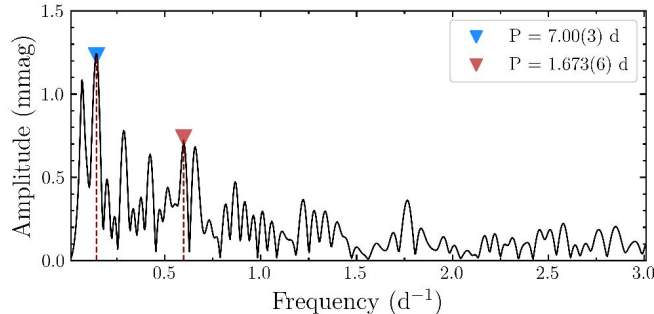
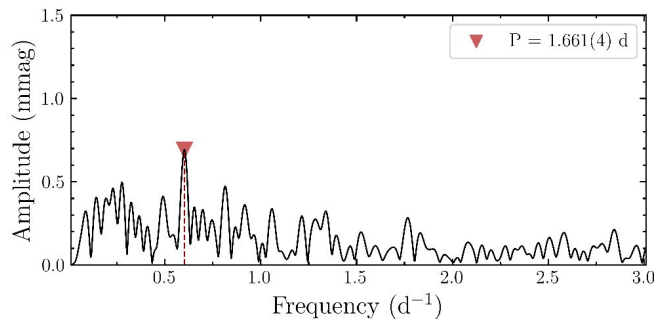
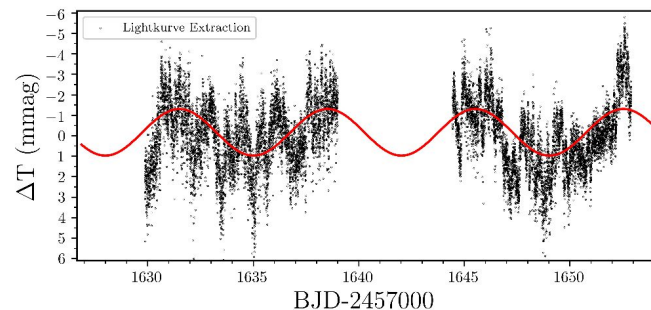
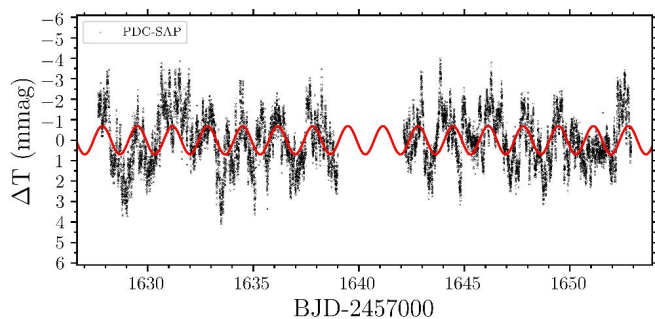
- Many *TESS* OB star studies have relied on 2-min pipeline provided PDC-SAP light curves.
- Using lightkurve v2.0¹ we aim to take advantage of 10/30 min FFI extraction and manual detrending.

Question: How does the *TESS* photometry of the Of?p stars HD 148937 ($P_{\text{rot}} = 7.03$ d) and LMC 164-2 ($P_{\text{rot}} = 7.96$ d) inform our search for rotational modulation in O-type stars?

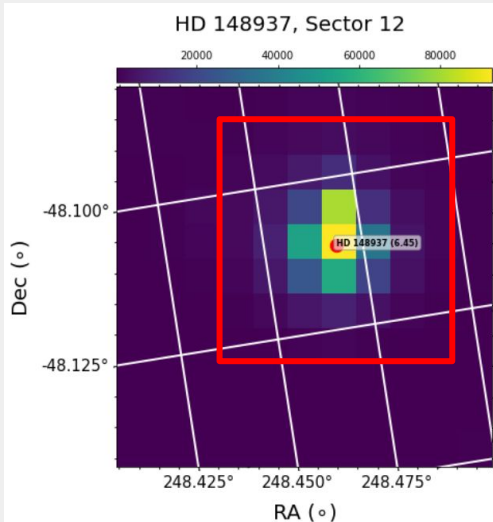
1. <https://docs.lightkurve.org/>

Case 1: HD 148937 - ($P_{\text{rot}} = 7.03$ d)

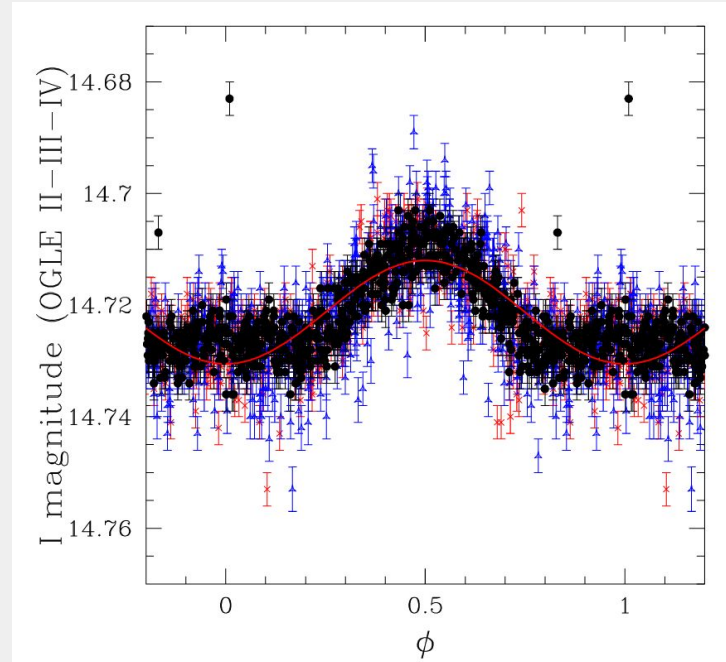
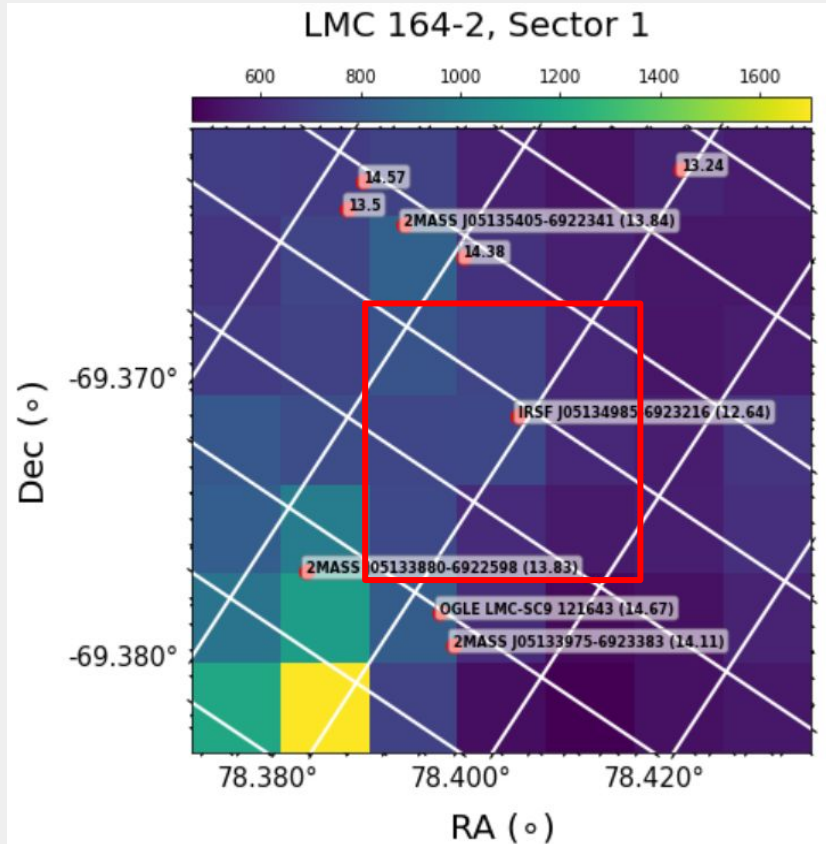
- O-O binary with a magnetic component (see talk by Abigail Frost).
- ~ 7.03 d period from H α equivalent width variations (Nazé et al. 2008, 2010 & Wade et al. 2012).



Wade et al. 2012, MNRAS, Vol. 419, 3, pp. 2459-2471.



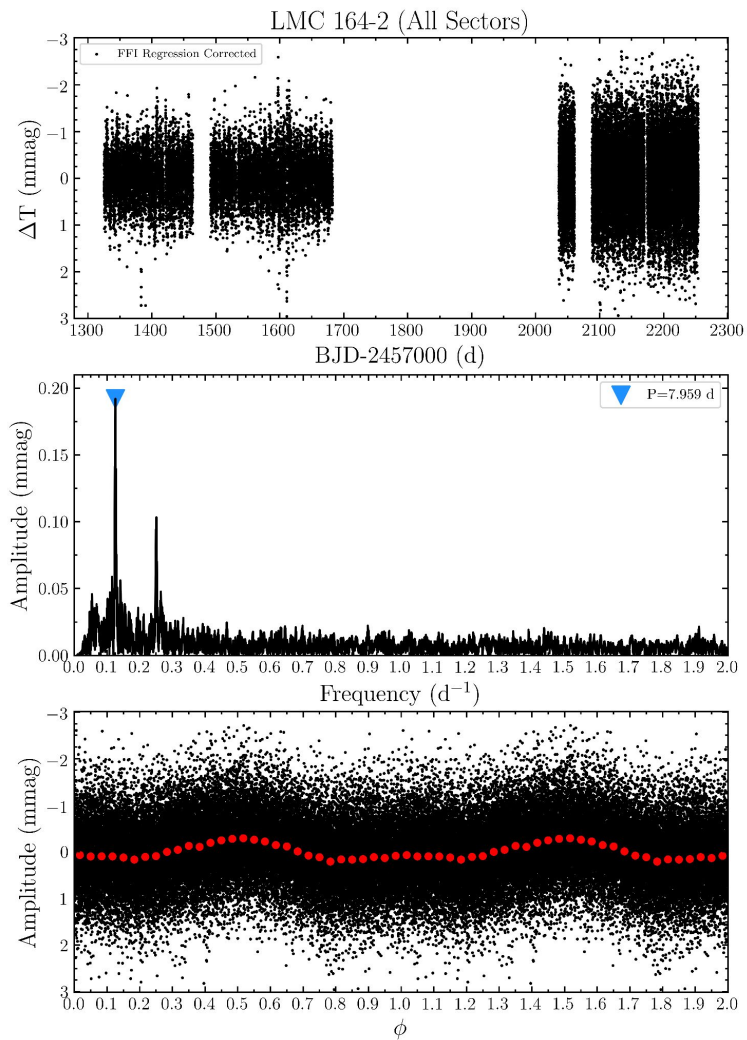
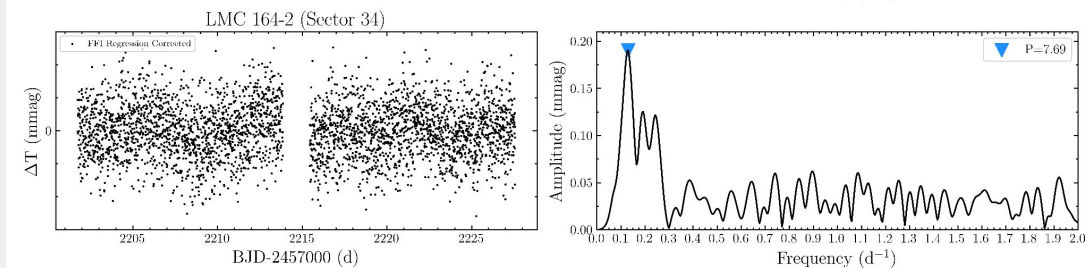
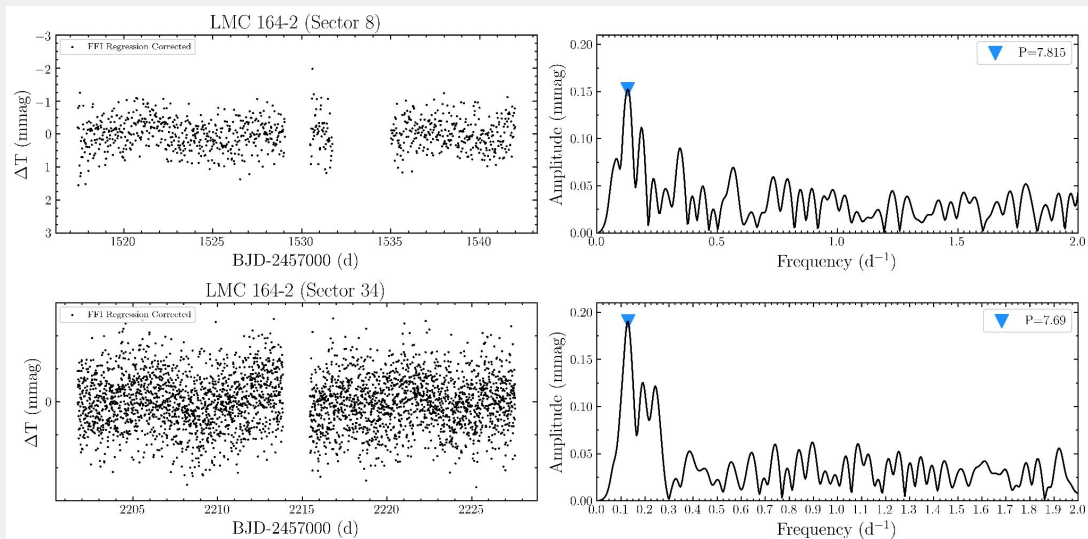
Case 2: LMC 164-2 - ($P_{\text{rot}} = 7.96$ d)



Nazé et al. 2015 A&A 577, A107

Case 2: LMC 164-2 - ($P_{\text{rot}} = 7.96$ d)

- Observed over 21 sectors in Cycles 1 and 3.
- Extraction from 30-min and 10-min FFIs using lightkurve 2.0



Conclusions:

Case 1: HD 148937

- *TESS* pipeline can remove O star rotational signals (~ 7 d) in PDC-SAP light curves (optimized for exoplanets).
- Recover 7.03 d spectroscopic period using a manual detrending process with lightkurve 2.0
- Recommend a manual detrending process to verify PDC-SAP light curves and check contamination / aperture.
- Scheduled to be observed again by *TESS* in June 2021.

Case 2: LMC 164-2

- Rotational signals ~ 8 d can be recovered from single sector (~ 27 d) 30/10-min FFIs.
- Precision improves with addition of multiple sectors.
- Potential to extend search for O star rotation in *TESS* photometry to Magellanic Clouds (continuous viewing zones).